Water Pump



Category: **Plumbing & Pipe Fitting** Type: **Full SOP** Priority: **High** 9 labs

Covers these lab items: Pump sets; Water Pump; WATER PUMP 1 PHASE; WATER PUMP 1 PHASE ISI Mark

What it is used for

A water pump moves water from a source to a required location through pipes or hoses. Correct installation, priming where required, and safe electrical operation help prevent dry running, leaks, and electric shock.

Main parts

- Electric motor supplies power to drive the pump.
- Pump casing contains the impeller or pumping mechanism.
- Inlet connection draws water into the pump.
- Outlet connection delivers water into the discharge pipe.
- Shaft and seals transfer motion while helping prevent leakage.
- Control switch or starter starts and stops the pump.
- Base, mounting feet, or supports hold the pump securely.

Before you start

1. Read the pump manual and confirm its voltage, capacity, and intended water use.
2. Ensure installation and electrical connections are handled by trained personnel.
3. Inspect the casing, pipes, hoses, seals, cable, plug, and switch for damage.
4. Place the pump on a stable, dry, secure base near suitable pipe connections.
5. Check that the inlet is clear and the required valves are correctly positioned.
6. Fill or prime the pump when required by the manufacturer's instructions.

How to use

1. Switch off and isolate the power supply before checking connections.
2. Connect the inlet and outlet pipes securely using compatible fittings and seals.
3. Open the required valves and confirm that the water source is available.
4. Complete priming or filling according to the pump design and manual.
5. Ensure the pump and surrounding electrical connections are dry before energising.
6. Switch on the pump using its approved control and observe the initial operation.
7. Check the water flow, discharge pressure if indicated, vibration, and visible leaks.
8. Stop the pump immediately if it runs dry, overheats, leaks heavily, or makes unusual noise.
9. Switch off and isolate power before clearing blockages or adjusting pipework.
10. After use, shut down the pump according to the manual and leave the area tidy.

⚠ Safety

- Never touch plugs, switches, or electrical connections with wet hands.
- Do not operate a pump with damaged wiring, exposed terminals, or waterlogged electrical parts.
- Never run a pump dry unless its design specifically permits it.
- Do not open the casing or touch internal parts while the pump is powered.
- Keep children away from the pump, moving parts, and electrical supply.
- Use only pumps approved for the intended water type; do not assume drinking-water suitability.

Care & storage

- Disconnect power before cleaning the exterior or checking accessible components.
- Inspect pipes, seals, fittings, and cables regularly for leaks and damage.
- Follow the manual for lubrication, priming, frost protection, and scheduled servicing.
- Report repeated tripping, overheating, reduced flow, or unusual vibration for authorised inspection.

Common mistakes to avoid

- Starting a pump without priming when priming is required.
- Running the pump with a blocked inlet or closed discharge valve when not permitted.
- Ignoring leaks, overheating, unusual vibration, or electrical faults.
- Handling electrical connections with wet hands or while power is connected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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